PHYS 7501:
Particles and Nuclei I

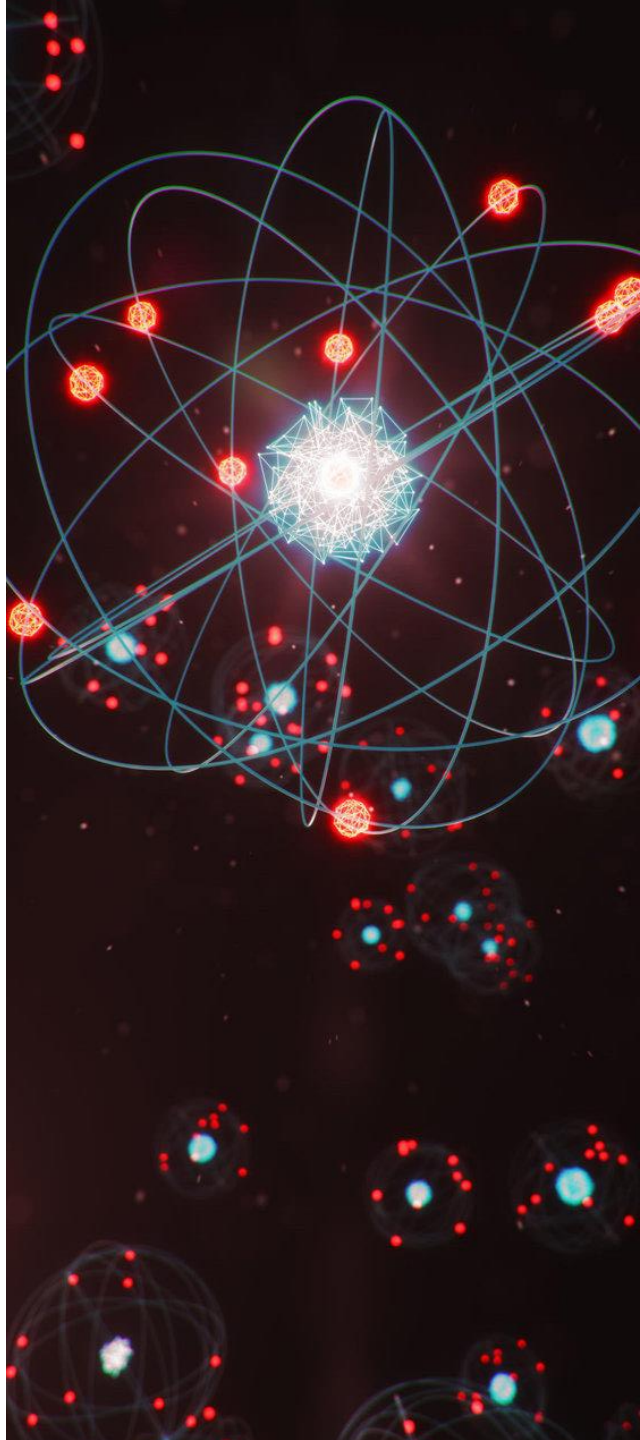
Tues./Thurs. 2 – 3:20PM
Clippinger 105

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Available via e-mail & Teams



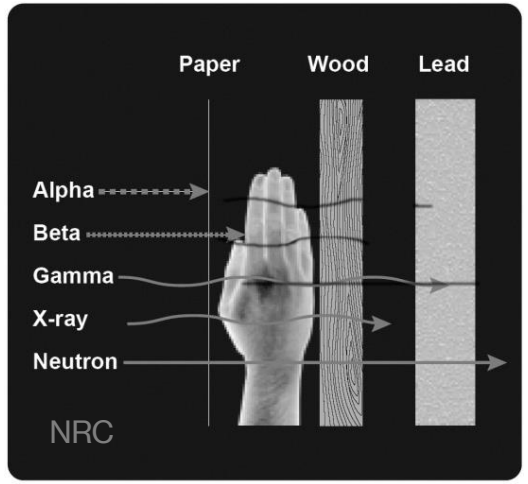
BETA DECAY

Modern Nuclear Chemistry, Chapter 8

Monday	Tuesday	Wednesday	Thursday	Friday
9/22	9/23	9/24	9/25	9/26
	Alpha Decay		**Beta Decay** HW #2 Due	
9/29	9/30	10/1	10/2	10/3
	Gamma Decay		**Fission**	
10/6	10/7	10/8	10/9	10/10
	Exam Day		**Nuclear Forces** HW#3 Due	

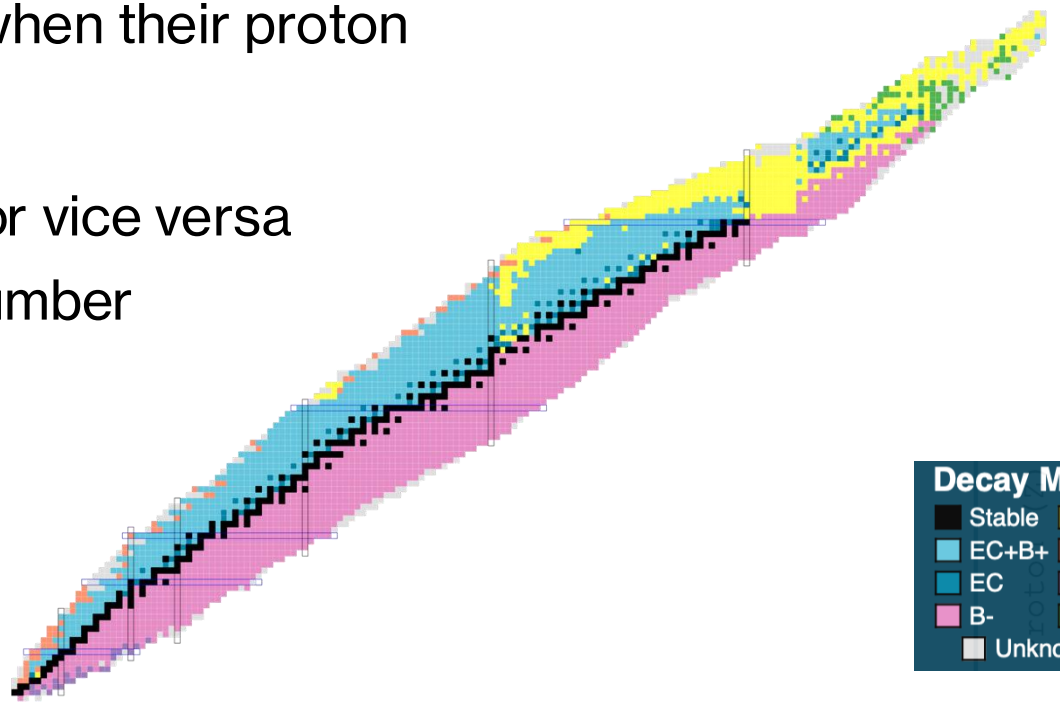
The Historical Perspective

- β decay was named for being the second most ionizing rays found to emanate from uranium samples
- Naturally occurring β rays were found to be fast-moving (negative) electrons but understanding the decay process was much more challenging
- Nuclei are unstable with respect to β decay when their proton and neutron numbers are out of balance
- β decay ”converts” a neutron into a proton or vice versa becoming more stable but maintains mass number



97% of radioactive nuclides β decay

So... it’s pretty important!



What is β decay? There are 3 types

$\bar{\nu}_e$ - anti-neutrino
ν_e- neutrino

Nucleus perspective

$\beta^-: {}^{A}_{Z}X_N \longrightarrow {}^{A}_{Z+1}Y_{N-1} + e^- + \bar{\nu}_e + Q_{\beta-}$

$\beta^+: {}^{A}_{Z}X_N \longrightarrow {}^{A}_{Z-1}Y_{N+1} + e^+ + \nu_e + Q_{\beta+}$

EC: ${}^{A}_{Z}X_N + e^- \longrightarrow {}^{A}_{Z-1}Y_{N+1} + \nu_e + Q_{EC}$

Nucleon perspective

$\beta^-: n \longrightarrow p + e^- + \bar{\nu}_e + Q_{\beta-}$

$\beta^+: p \longrightarrow n + e^+ + \nu_e + Q_{\beta+}$

EC: $p + e^- \longrightarrow n + \nu_e + Q_{EC}$

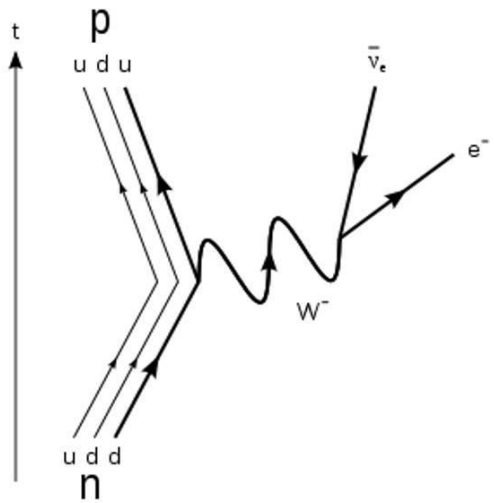
Quark perspective

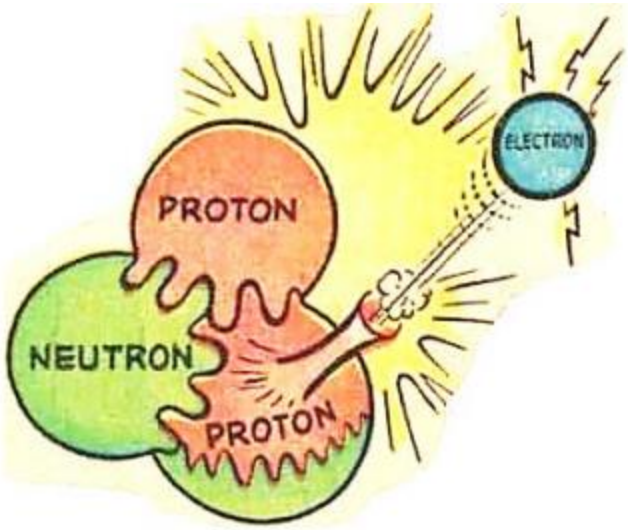
$\beta^-: d \longrightarrow u + e^- + \bar{\nu}_e + Q_{\beta-}$

$\beta^+: u \longrightarrow d + e^+ + \nu_e + Q_{\beta+}$

EC: $u + e^- \longrightarrow d + \nu_e + Q_{EC}$

- More complicated than α decay
  - Three-body final state
- β decays are mediated by the *weak interaction*
  - W^+, W^-, Z^0 bosons are the mediators
  - Responsible for transmutation of neutron into a proton plus an electron and an electron anti-neutrino...





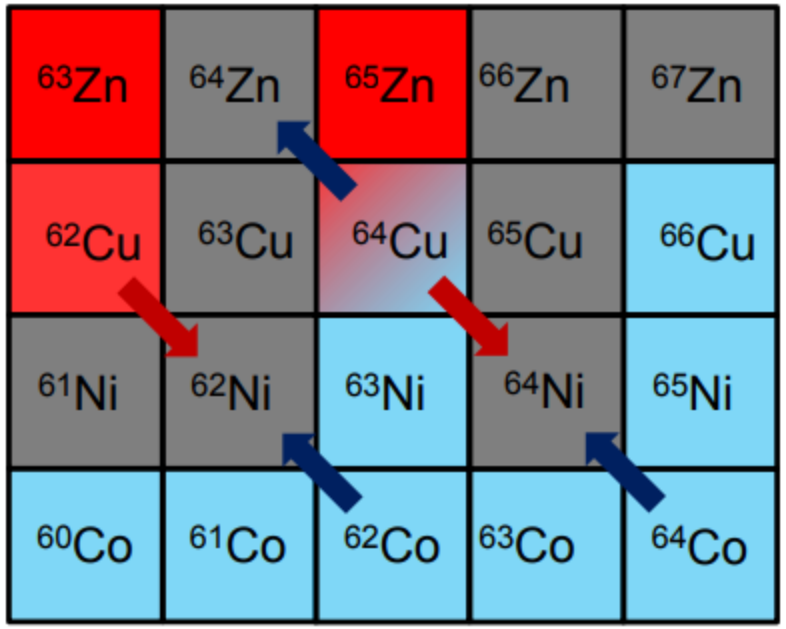
What does this look like on the Nuclear Chart?

$\beta^-: {}^{A}_{Z}X_N \longrightarrow {}^{A}_{Z+1}Y_{N-1} + e^- + \bar{\nu}_e + Q_{\beta-}$

$\beta^+: {}^{A}_{Z}X_N \longrightarrow {}^{A}_{Z-1}Y_{N+1} + e^+ + \nu_e + Q_{\beta+}$

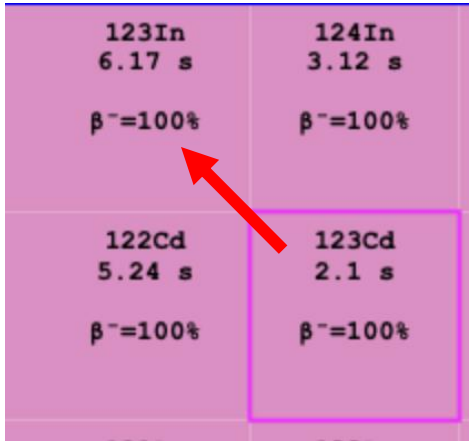
$\text{EC}: {}^{A}_{Z}X_N + e^- \longrightarrow {}^{A}_{Z-1}Y_{N+1} + \nu_e + Q_{\text{EC}}$

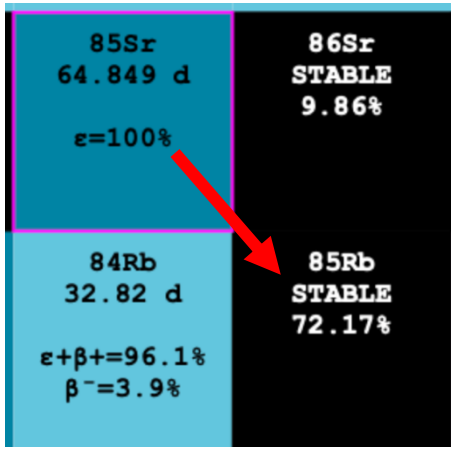
- Neutron-rich nuclei undergo β^- decay
- Proton-rich nuclei undergo β^+ decay
- Some nuclei can undergo both
- β decay can occur when $Q_\beta > 0$
- Preferred decay mode for most nuclei
  - Example:
    - $\beta^-: {}^{64}_{27}Co_{37} \longrightarrow {}^{64}_{28}Ni_{26} + e^- + \bar{\nu}_e + Q_{\beta-}$
    - $\beta^+: {}^{62}_{29}\text{Cu}_{33} \longrightarrow {}^{62}_{28}\text{Ni}_{34} + e^+ + \nu_e + Q_{\beta+}$



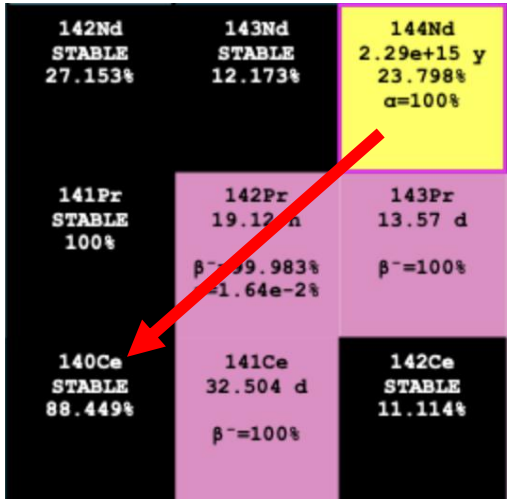
Explore the NNDC

- What kind of decay do the following nuclei undergo? Write out the decay process:
  - ^{123}Cd $^{123}Cd \rightarrow {}^{123}In + e^{-} + \overline{\nu_e}$
  - ^{85}Sr $^{85}Sr \rightarrow {}^{85}Rb + e^{+} + \nu_e$
  - ^{126}I $^{126}I \rightarrow {}^{126}Xe + e^{-} + \overline{\nu_e}$ $^{126}I \rightarrow {}^{126}Te + e^{+} + \nu_e$
  - ^{144}Nd $^{144}Nd \rightarrow {}^{140}Ce + \alpha$









β-Decay Energetics

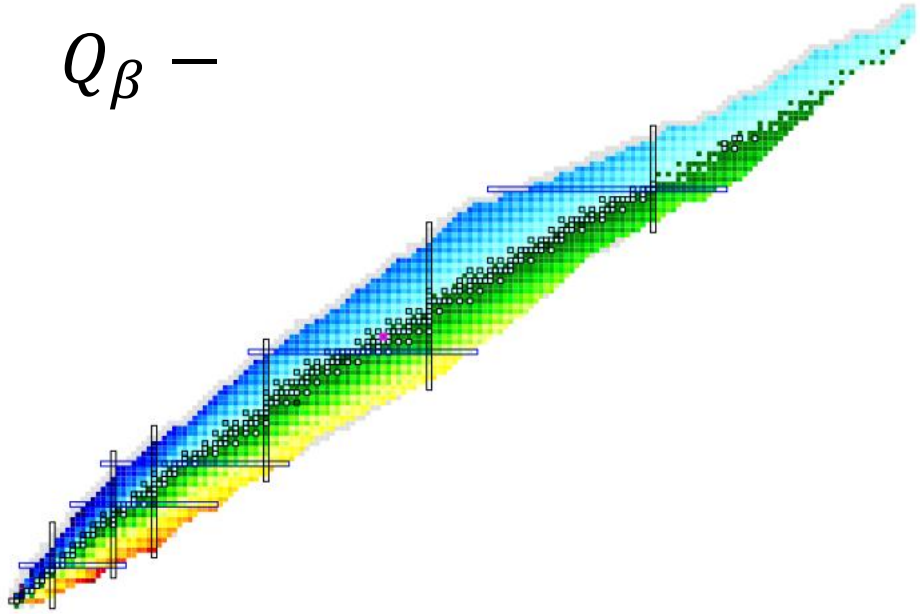
$$Q_{\beta-} = \Delta(Z, A) - \Delta(Z+1, A)$$

$$Q_{\beta+} = \Delta(Z, A) - \Delta(Z-1, A) - 2\mathrm{m_e}c^2$$

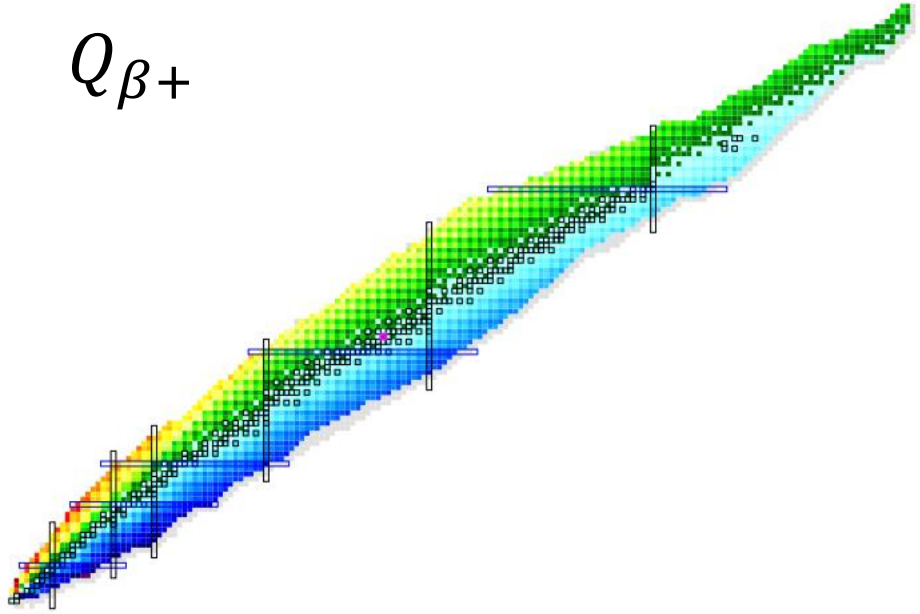
$$Q_{\mathrm{EC}} = \Delta(Z, A) - \Delta(Z-1, A)$$

Atomic mass excesses have the electron masses included, so it's only for positron emission that we have to take into account that we will not be gaining an electron with the atom, but instead effectively losing one.

$Q_{\beta}-$

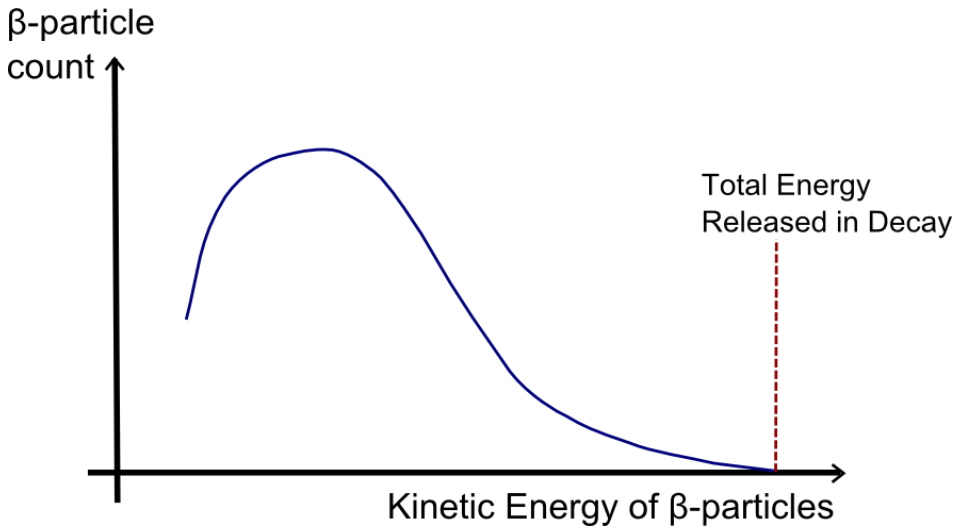


$Q_{\beta+}$



β-Decay Energy Spectrum

- Early experiments investigating the "β ray" showed that it was not emitted with a singular energy, like the "α ray", but rather in a continuum of energies
  - The maximum energy is the decay Q value
- Furthermore, the reaction $n \rightarrow p + e^-$ doesn't conserve spin!
  - $J_n = J_p = J_e = \frac{1}{2}$..... So, $0 \leq J_P + J_e \leq 1 \neq \frac{1}{2}$

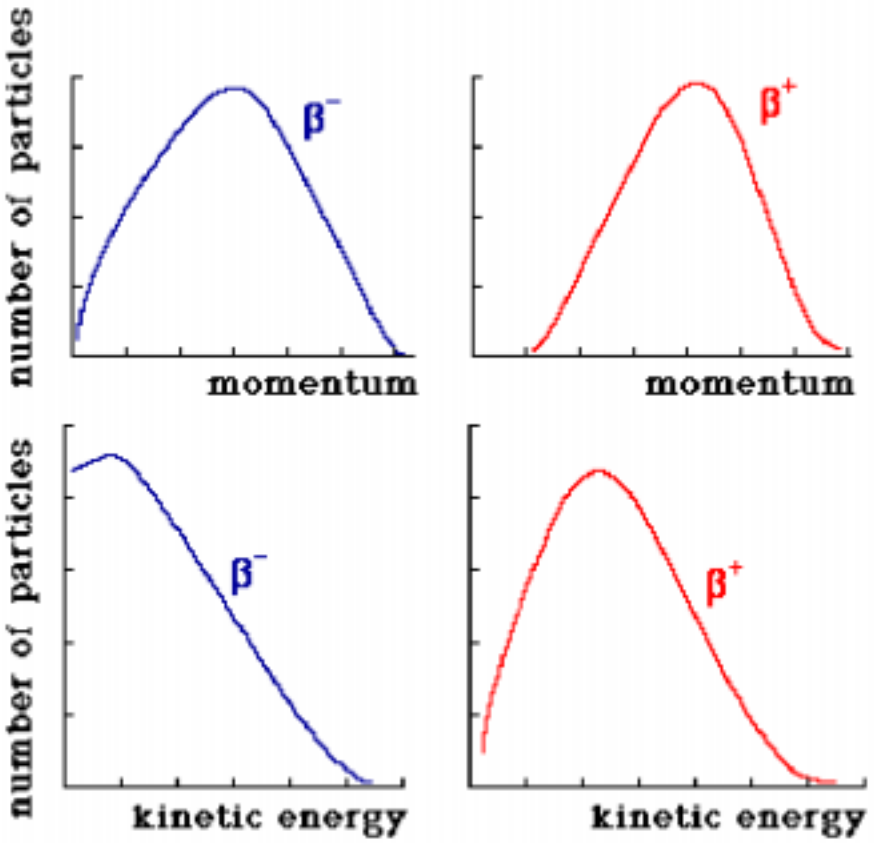


- Pauli postulated the existence of a third hypothetical particle (the neutrino) to explain the shape of the β-decay spectrum.
  - He proposed this in a letter to Lise Meitner
- Given the considerations, it was postulated that υ was a spin-1/2 particle (fermion) that is massless (true enough for our purposes) and electrically neutral
- In one of his last works before switching to experimental work, Fermi postulated that nucleons could act as sources and sinks of neutrinos, which was the only successful theory of interactions between quantum particles (in 1934)

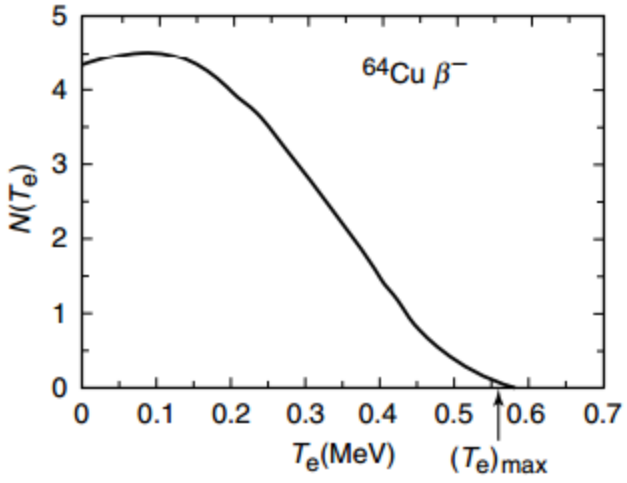
We know this is true today, but *Nature* actually rejected Fermi's paper for being "too remote from physical reality"

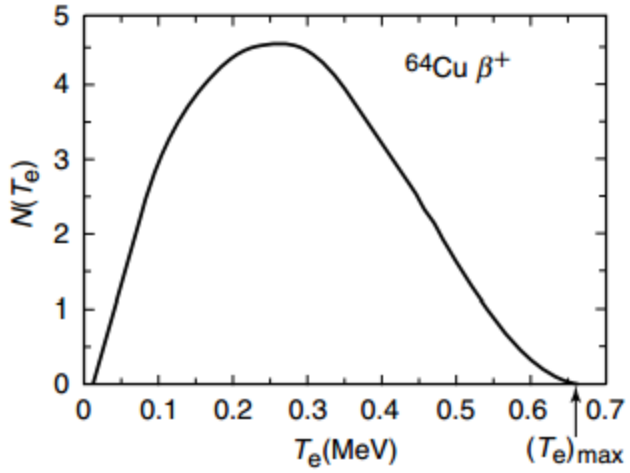
β-Decay Energy Spectrum

- Three-body decay
- Continuous energy distribution from some maximum down to zero at the Q value
- Energy split between participants in the decay
  - Daughter, β, and neutrino



Physics LibreTexts





Fermi Theory of β Decay

- Fermi posited that a decay-causing interaction was a weak perturbation
- The decay constant in this case would be

$$\lambda = \frac{2\pi}{\hbar}\left|\int \Psi^*_{\text{final}}\, \mathrm{V_P} \Psi_{\text{initial}} \mathrm{d}\tau\right|^2 \rho(E_f)$$

This is known as Fermi's Golden Rule

- The wave functions $\boldsymbol{\Psi}$ represent the complete initial and final states of the entire system, $\boldsymbol{V_P}$ is a very small perturbative interaction that simulates the transition, and $\boldsymbol{\rho(E_f)}$ is the density of quantum mechanical states available after the transition (often written as $\frac{\boldsymbol{dn}}{\boldsymbol{dE}}$)
- $\boldsymbol{\Psi_{initial}}$ contains only the parent nucleus, but $\boldsymbol{\Psi^*_{final}}$ has contributions from all resultant particles
- For β decay, $\boldsymbol{\Psi_{initial} = \phi_{gs}(^A Z)}$. This is the parent in the ground state
- The final wavefunction has three parts: $\boldsymbol{\Psi^*_{final} = \phi^*_j(^A Z)\phi*(e)\,\phi*(\upsilon)}$
  - daughter nucleus in the appropriate final state j,
  - traveling wave of the electron,
  - traveling wave of the neutrino,
  - energy and angular momentum must be conserved

- The solution for $\lambda = \frac{2\pi}{\hbar}\left|\int \Psi^*_{\text{final}}\, \mathrm{V_P} \Psi_{\text{initial}} \mathrm{d}\tau\right|^2 \rho(E_f)$ is split into two parts
  - **Determination of $\boldsymbol{\rho(E_f) = \frac{dn}{dE}}$ using quantum statistical mechanics**
    - # of ways that the electron and (anti)neutrino can share the decay energy E_f
  - **Matrix element $\left|\int \Psi^*_{final} V_P \Psi_{initial} d\tau\right|^2$ describing the interaction**
    - Description of initial state *i* to one of many possible final states *f*
    - Wave-function overlap between initial and final states (requires theory)

Determination of $\boldsymbol{\rho(E_f)}$:

- How many ways can E_f be split between $\mathrm{e^-}$ and υ?
- The number of final states is

$$dn = (\#\ electrons\ in\ volume\ V\ of\ momentum\ space\ with\ momentum\ between\ p_e\ and\ p_e + dp_e) \times (\#\ neutrinos\ in\ volume\ V\ of\ momentum\ space\ with\ momentum\ between\ p_\nu\ and\ p_\nu + dp_\nu)$$

- Can actually solve for this using the volume of a spherical shell in phase space where the unit cell is h^3

- Then, the number of states is $\boldsymbol{dn = \frac{V4\pi p_X^2 dp_X}{h^3}}$ but we need this expression for both e^- and υ

- So,

$$\boldsymbol{dn = \frac{16\pi^2 V^2 p_e^2 p_\upsilon^2 dp_e dp_\upsilon}{h^6}}$$

- For $\boldsymbol{m_\nu = 0}$, $\boldsymbol{p_\upsilon = \frac{T_\upsilon}{c} = \frac{Q - T_e}{c}}$. Then $\boldsymbol{dp_\upsilon = \frac{dQ}{c}}$

- Substituting all of this back in, we get

$$\boldsymbol{dn = \frac{16\pi^2 V^2}{h^6 c^3}(Q - T_e)^2 p_e^2 dp_e dQ}$$

- Meaning that the change in number of final states for a change in Q is:

$$\boldsymbol{\frac{dn}{dQ} = \frac{16\pi^2 V^2}{h^6 c^3}(Q - T_e)^2 p_e^2 dp_e}$$

This factor represents the phase space of the decay, which is why the β^- and β^+ spectra look differently

This is $\boldsymbol{\rho(E_f)}$ that we needed!

Matrix element $\left|\int \Psi^*_{final} V_P \Psi_{initial} d\tau\right|^2$ describing the interaction

- Treat both the electron and (anti)neutrino wave functions as radial plane waves

$$\boldsymbol{\phi_e = Ae^{ik_e r} = \frac{1}{\sqrt{V}} e^{ik_e r}}$$

$$\boldsymbol{\phi_v = Be^{ik_v r} = \frac{1}{\sqrt{V}} e^{ik_v r}}$$

- We need to apply normalization conditions to obtain ***A*** and ***B***.
- If you expand the exponentials around $\boldsymbol{r \sim 0}$ (the nuclear volume), then $e^{ikr} = 1 + ikr + \cdots \approx 1$ (Taylor Expansion) and $\boldsymbol{\phi(r \sim 0) \approx \frac{1}{V}}$.
- The probability for emitting an electron with momentum between $\boldsymbol{p_e}$ and $\boldsymbol{dp_e}$ is

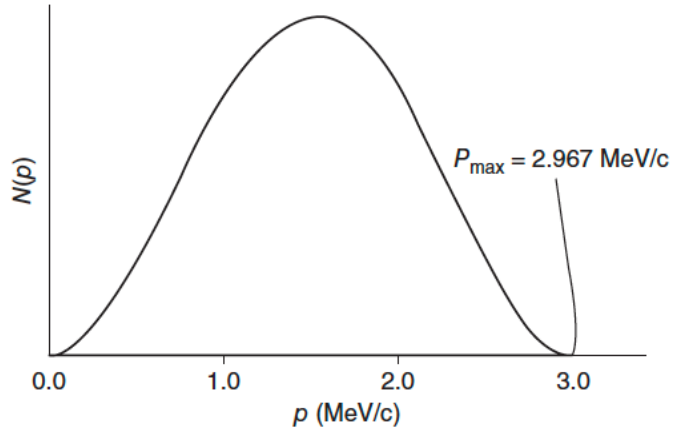
$$\boldsymbol{\lambda(p_e)dp_e = \frac{1}{2\pi^3 \hbar^7 c^3} \left|M_{fi}\right|^2 g^2 (Q - T_e)^2 p_e^2 dp_e}$$

g is the perturbation constant

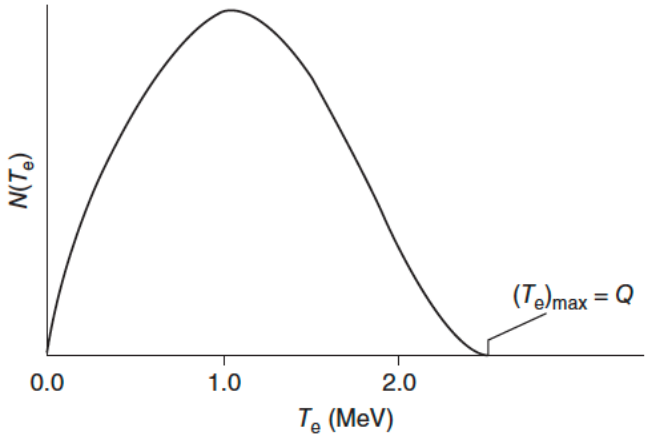
$\left|M_{fi}\right|$ is the nuclear matrix element representing the overlap between the initial and final nuclear states. It should be evaluated with detailed wavefunctions such as from the Shell Model

$$\lambda(p_e)dp_e = \frac{1}{2\pi^3\hbar^7c^3}\left|M_{fi}\right|^2 g^2(Q-T_e)^2 p_e^2 dp_e$$

- This form clearly goes to zero at $p_e = 0$ and also at $T_e = Q$ with a maximum in between!



- This exactly describes the shape of the β-decay spectrum



- One aside: we do technically need to consider Coulomb distortion (daughter repels e^+ and attracts e^-) in this spectrum which we do with a scaling factor known as the **Fermi Function**, $F(Z_{daughter}, p_e)$. This is hard to calculate so we use tabulated values.

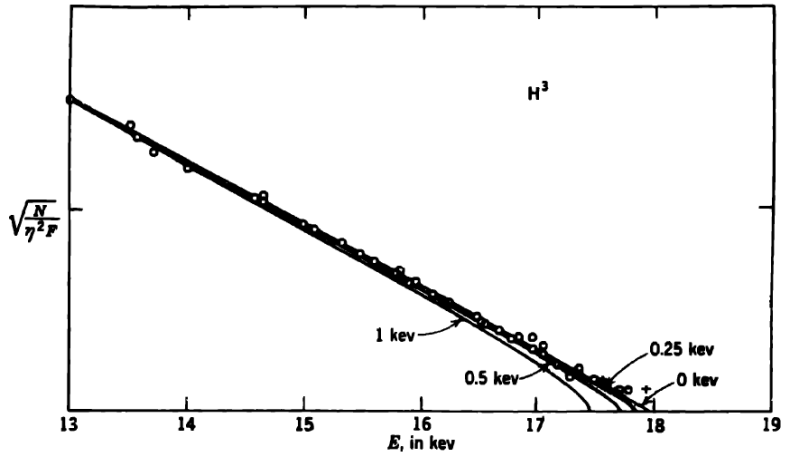
β-decay rate per electron momentum

- If we then combine all pieces of information,

$$\lambda(p_e)dp_e = \frac{|M_{fi}|^2}{2\pi^3\hbar^7c^3} F(Z_d, p_e)\,(Q - T_e)^2 \left(1 - \frac{m_\nu^2 c^4}{(Q-T_e)^2}\right)^{1/2} p_e^2 dp_e$$

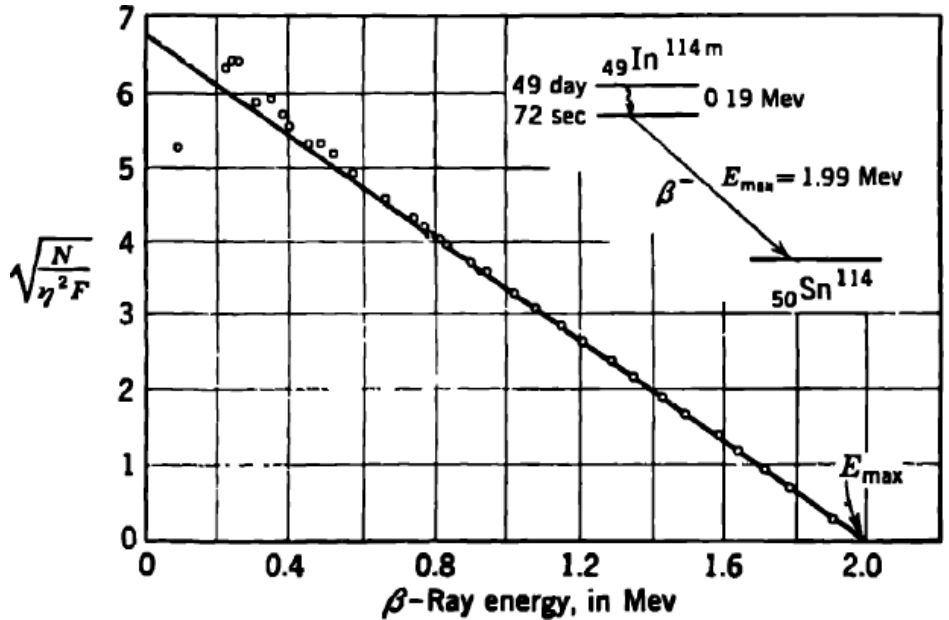
- From this, we can see that

$$\left(\frac{\lambda(\mathrm{p_e})}{\mathrm{p_e^2} F(Z_d, p_e)}\right)^{1/2} \propto |M_{fi}|^2 (Q - T_e)^2$$



R. Evans, The Atomic Nucleus (1955)

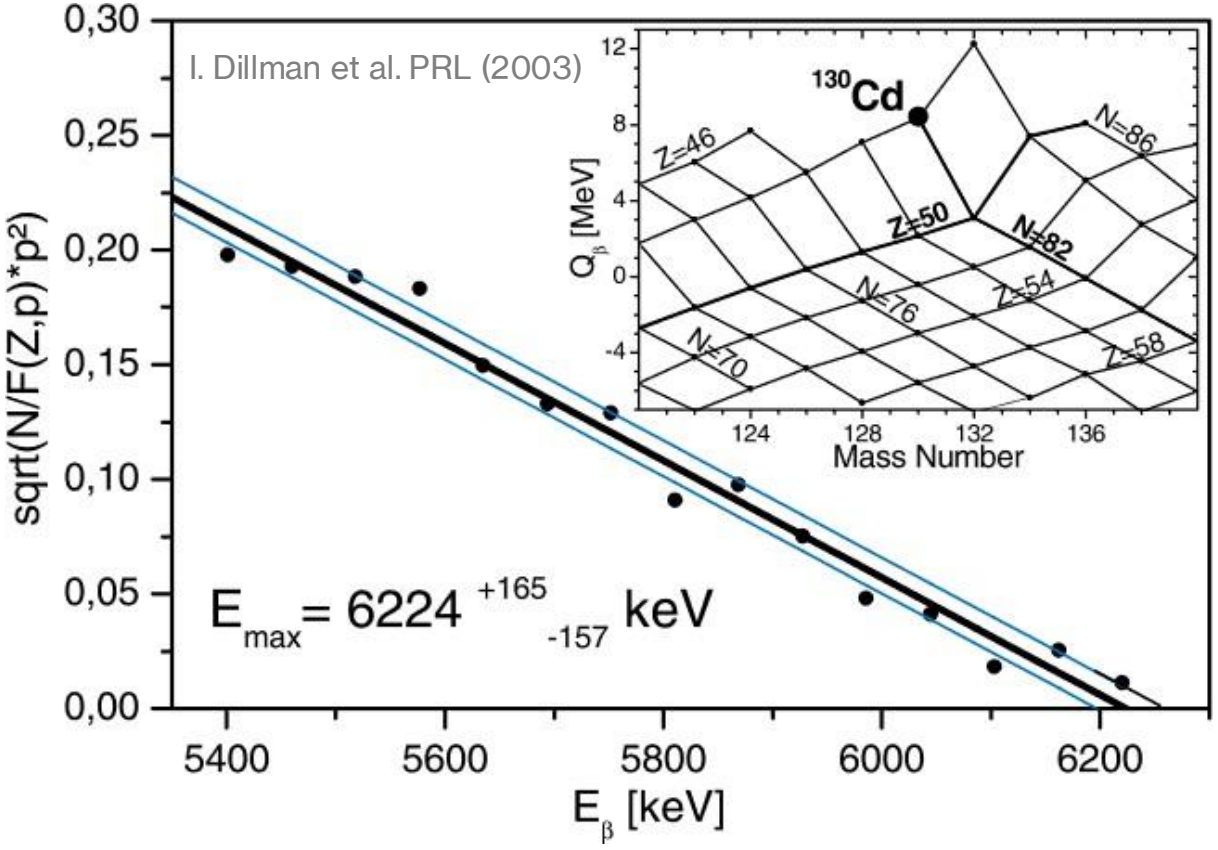
- This yields a straight line at $Q = T_e$ that intersects the horizontal axis
- This is known as a **Kurie Plot**!
  - This only applies to *allowed* transitions

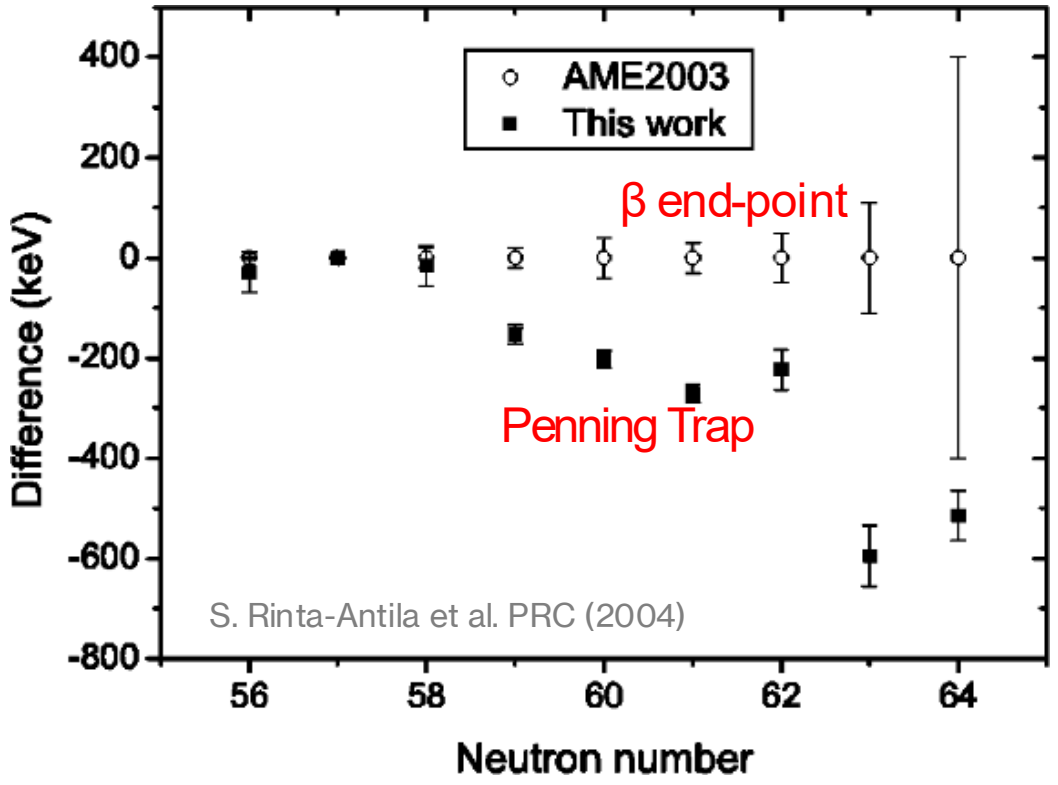


Aside: nuclear mass from the Kurie plot

- Of course, the end-point of the Kurie plot can also be used to obtain nuclear masses, assuming the mass of the less-exotic nucleus in the decay is known
- This is known as the "β end-point method"

...which has fallen out of favor lately due to systematic discrepancies with higher-precision techniques





Total β-decay rate and *ft* values

- Integrating the form of the b-decay spectrum over all electron momenta yields the decay constant:

$$\lambda = \frac{g^2 |M_{fi}|^2}{2\pi^3 \hbar^7 c^3} \int_0^{p_{max}} F(Z_d, p_e)\, p_e^2 (Q - T_e)^2 dp_e$$

Assuming a constant NME

- So this is hard… but we can define the Fermi Integral and use numerical methods to integrate
- Dimensionless Fermi Integral:

$$f(Z_d, Q) \propto \int_0^{p_{max}} F(Z_d, p_e)\, p_e^2 (Q - T_e)^2 dp_e$$

- Then, our expression for the decay rate becomes

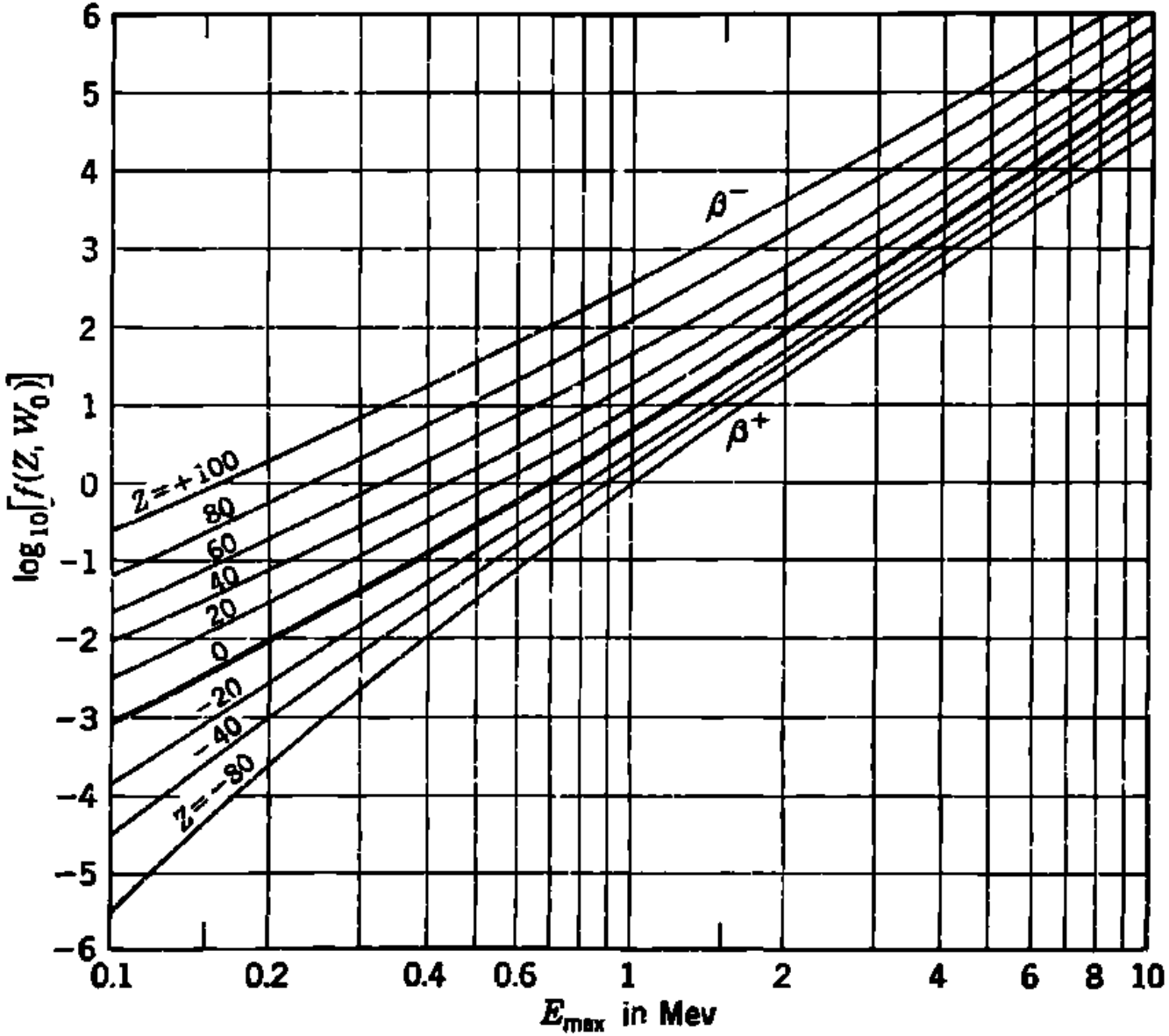
$$\lambda = \frac{g^2 |M_{fi}|^2}{2\pi^3 \hbar^7 c^3} f(Z_d, Q) = \frac{ln(2)}{t_{1/2}}$$

The Fermi Integral

- In case you find yourself needing to calculate the $f(Z_d, Q)$ use a graph such as this one, which was determined numerically.

- Could instead reproduce the calculations of Towner & Hardy PRC, 015501 (2015)

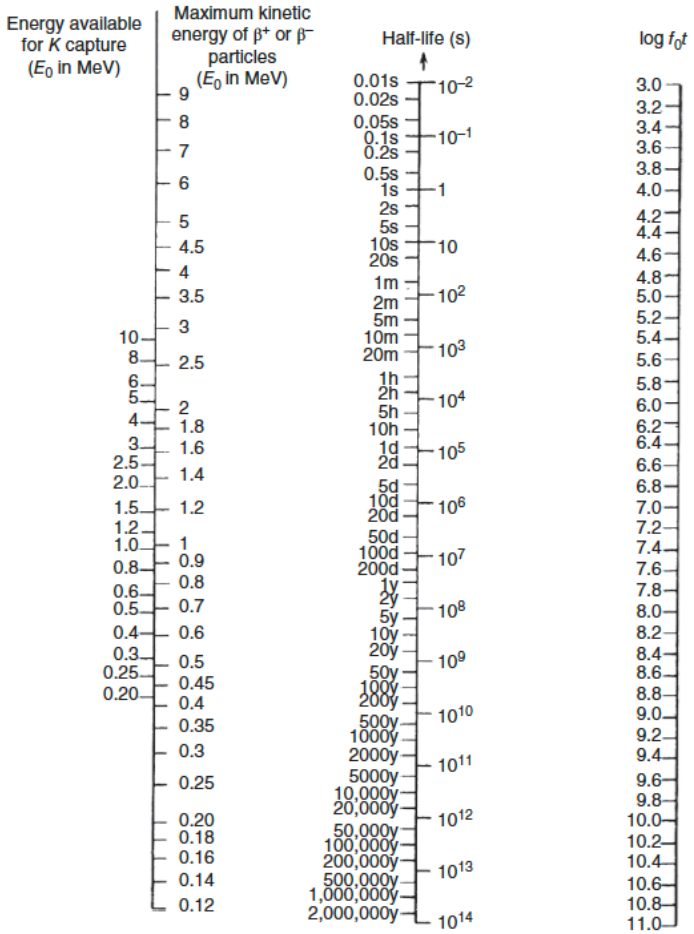
R. Evans, The Atomic Nucleus (1955)



- We can then write out the *comparative half-life* or *ft value* using

$$ft_{1/2} = ln(2)\frac{2\pi^3\hbar^7 c^3}{g^2\left|M_{fi}\right|^2 m_e^5 c^4}$$

- Can be easily measured in experiments and should really only depend on the matrix element
- β-decay half-lives span many orders of magnitude so the *ft* values will too. We typically use $log(ft)$ to characterize observed β-decays.



Fermi vs. Gamow-Teller Decays

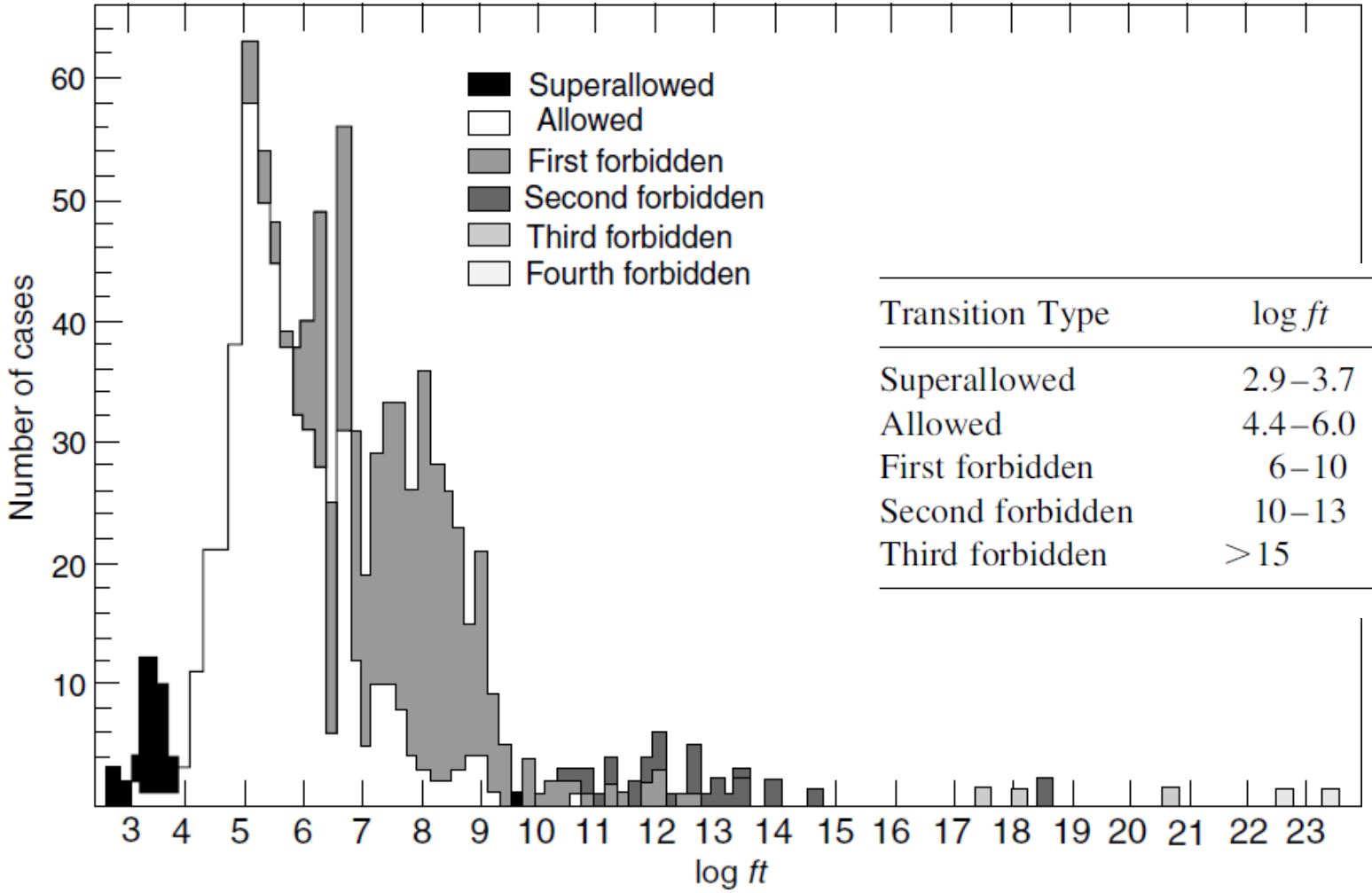
- Earlier we made the approximation for the wave functions that the Taylor expansion went to 1. That essentially made the result that β decay would result in no angular momentum transfer
- Yet, some observed transitions do have $\Delta J = 1$. These are Gamow-Teller decays that occur due to the coupling of the spin between the electron and neutrino, which can be anti-aligned $(S = 0)$ or aligned $(S = 1)$
- For the **anti-aligned case**, zero angular momentum is carried away, and parity is conserved. These are **Fermi decays** that we've covered so far
- For the **aligned case**, 0 or 1 units of angular momentum can be carried away, and parity is still conserved. These are **Gamow-Teller decays**
  - GT transitions cannot happen between 0^+ to 0^+ states
- Generally 0^+ to 0^+ transitions are called "super allowed" where the initial and final state wave functions are nearly identical. These can only be Fermi in nature.

Decay Selection Rules

- Ignoring higher order terms in the Taylor expansion omits the possibility of angular momentum transfer.
- If the angular momentum transfer does occur, higher-order terms need to be included, and the matrix element can no longer be independent of p_e
- For cases where $\Delta \mathrm{J} > 1$ and/or $\Delta\pi = \mathrm{yes}$, the leading order overlap $\left|\boldsymbol{M_{fi}}\right|^{\mathbf{2}} = 0$ so a higher-order term is needed
- This combined with if there is or is not a change in parity is referred to as how *forbidden* a decay is... though it's not really forbidden
  - **Super−Allowed :** $\mathbf{0^+ \rightarrow 0^+}$
  - **Allowed:** $\mathbf{0^+ \rightarrow 1^+}$ **or** $\boldsymbol{\Delta J = 0\ or\ 1}$ **and** $\boldsymbol{\Delta\pi = no}$
  - **First Forbidden:** $\boldsymbol{\Delta J = 0\ or\ 1}$ **and** $\boldsymbol{\Delta\pi = yes}$
  - **Second Forbidden:** $\boldsymbol{\Delta J = 2}$**and** $\boldsymbol{\Delta\pi = no}$

Empirical log(ft) Values and Selection Rules

W. Meyerhof, Elements of Nuclear Physics (1967)

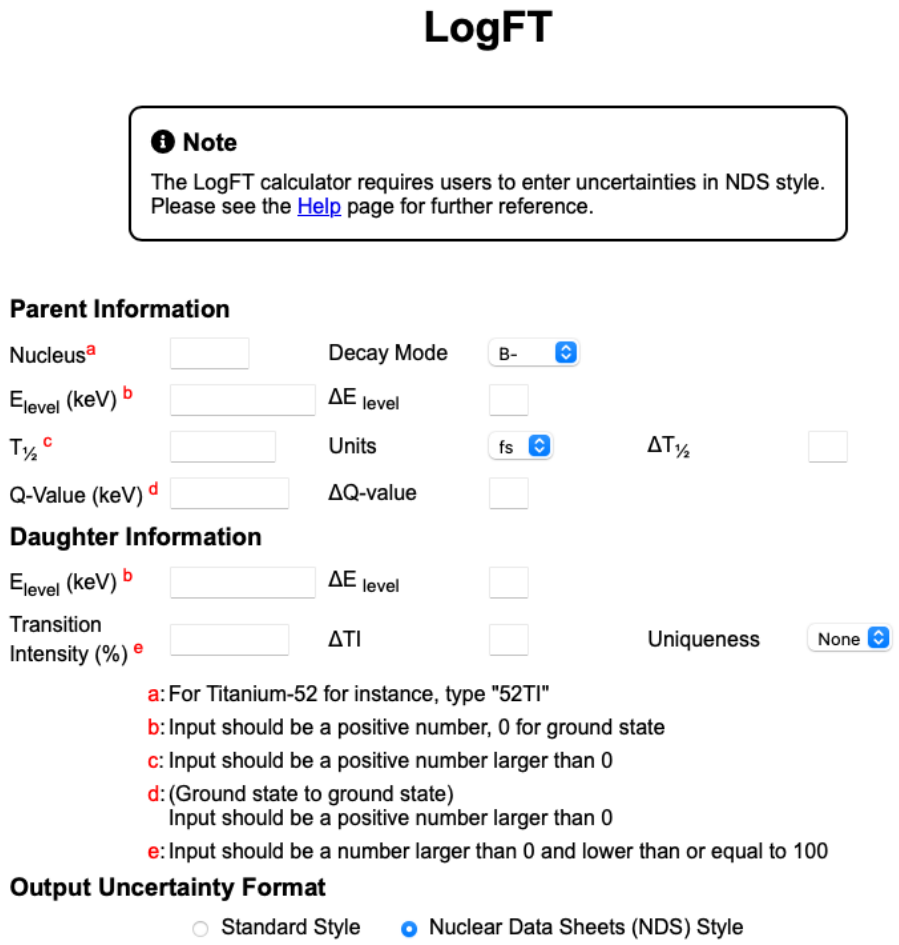


Loveland, Morrissey, & Seaborg, Modern Nuclear Chemistry (2006)

Transition Type	log *ft*	L_β	$\Delta\pi$	Fermi ΔI	Gamow–Teller ΔI
Superallowed	2.9–3.7	0	No	0	0
Allowed	4.4–6.0	0	No	0	0, 1
First forbidden	6–10	1	Yes	0, 1	0, 1, 2
Second forbidden	10–13	2	No	1, 2	1, 2, 3
Third forbidden	>15	3	Yes	2, 3	2, 3, 4

NNDC Online Calculator for *ft* Values

- If you already have experimental information about the decay such as the Q-value and half-life, you can calculate the *ft* value



LogFT

Note

The LogFT calculator requires users to enter uncertainties in NDS style. Please see the Help page for further reference.

Parent Information

Nucleus[a] | Decay Mode: B-

E_{level} (keV) [b] | ΔE_{level}

$T_{½}$ [c] | Units: fs | $\Delta T_{½}$

Q-Value (keV) [d] | ΔQ-value

Daughter Information

E_{level} (keV) [b] | ΔE_{level}

Transition Intensity (%) [e] | ΔTI | Uniqueness: None

a: For Titanium-52 for instance, type "52TI"

b: Input should be a positive number, 0 for ground state

c: Input should be a positive number larger than 0

d: (Ground state to ground state)
Input should be a positive number larger than 0

e: Input should be a number larger than 0 and lower than or equal to 100

Output Uncertainty Format

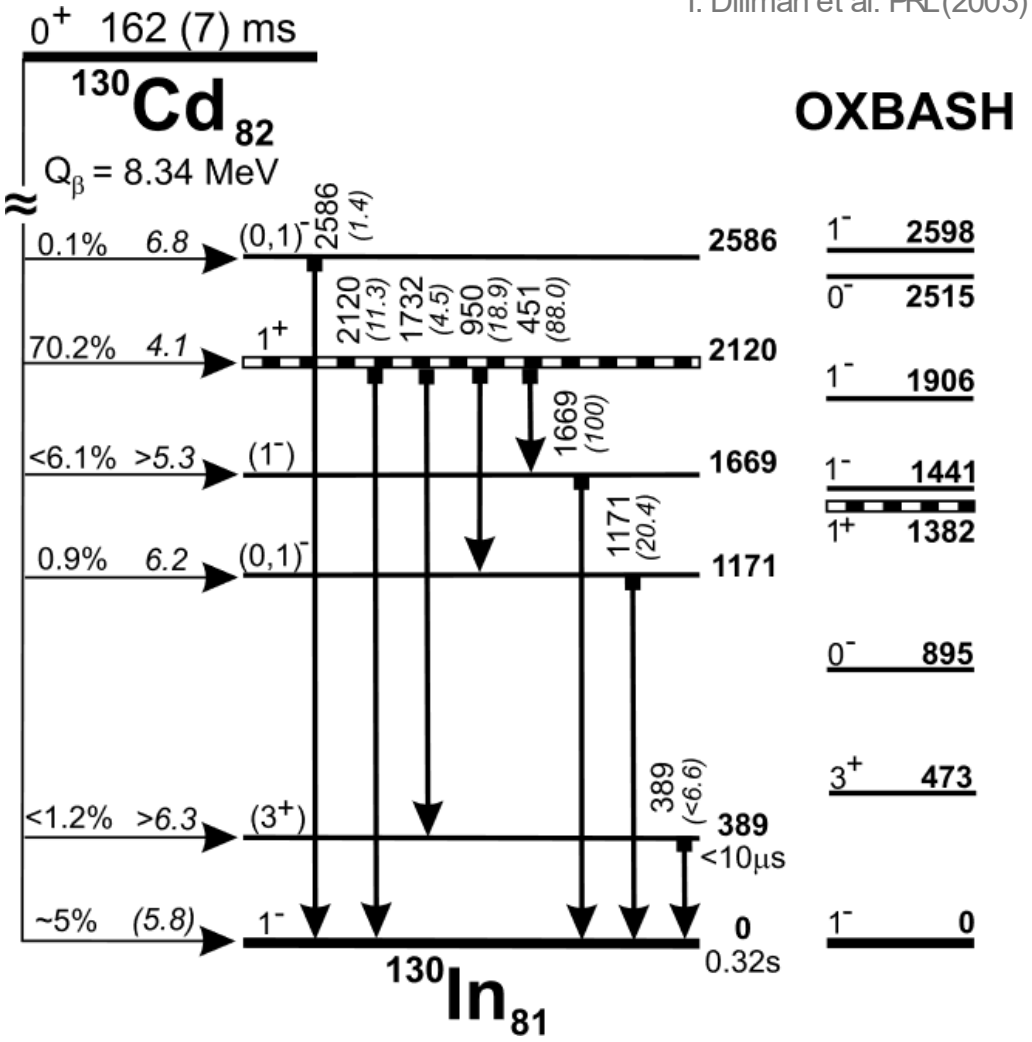
○ Standard Style ◉ Nuclear Data Sheets (NDS) Style

Using *ft* for Spectroscopy

- We established that the *ft* value is linked to a particular $\Delta J, \Delta\pi$, the measurement of an *ft* can constrain the J^{π} of an unknown state if the β decay proceeds from a state of known J^{π}
- Nuclear masses combined with γ-ray measurements provide the transition Q-value (*f*)
- The % of time that a decay proceeds through a given state is the branching ratio *(t)*

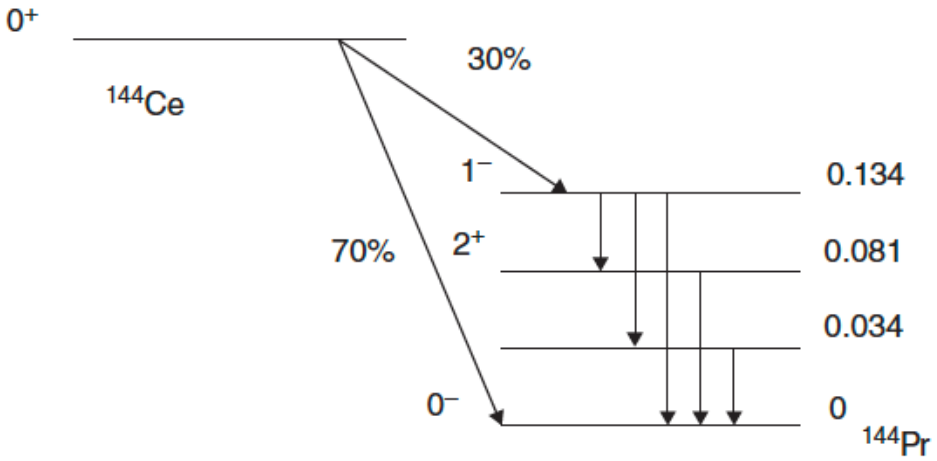
Transition Type	log *ft*	L_β	$\Delta\pi$	Fermi ΔI	Gamow–Teller ΔI
Superallowed	2.9–3.7	0	No	0	0
Allowed	4.4–6.0	0	No	0	0, 1
First forbidden	6–10	1	Yes	0, 1	0, 1, 2
Second forbidden	10–13	2	No	1, 2	1, 2, 3
Third forbidden	>15	3	Yes	2, 3	2, 3, 4

I. Dillman et al. PRL (2003)



Try a Classifying a Few Decays

- **^{33}Mg has a ground state J^{π} of $\frac{3}{2}^{-}$, what are the allowed decay J^{π}?**
  - Allowed: $\frac{1}{2}^{-}$, $\frac{3}{2}^{-}$, $\frac{5}{2}^{-}$
- **^{60}Co has a ground state J^{π} of 5^{+}, what are the allowed decay J^{π}?**
  - Allowed: $4^{+}, 5^{+}, 6^{+}$
- A fake decay of ^{144}Ce is shown:



Why is there no decay to the 2^{+} level?

The ΔJ would be 2, and there is no change in parity making it a second forbidden transition

Classify These Decays

Parent G.S. J^{π}	Daughter J^{π}	Type	Fermi or GT?
0^+	2^-	First Forbidden	GT
$5/2^+$	$3/2^+$	Allowed	Fermi or GT (mixed)
1^-	3^+	Third Forbidden	GT
0^+	0^+	Superallowed Or Allowed	Fermi or GT

Transition Type	log *ft*	L_β	$\Delta\pi$	Fermi ΔI	Gamow–Teller ΔI
Superallowed	2.9–3.7	0	No	0	0
Allowed	4.4–6.0	0	No	0	0, 1
First forbidden	6–10	1	Yes	0, 1	0, 1, 2
Second forbidden	10–13	2	No	1, 2	1, 2, 3
Third forbidden	>15	3	Yes	2, 3	2, 3, 4

Electron Capture

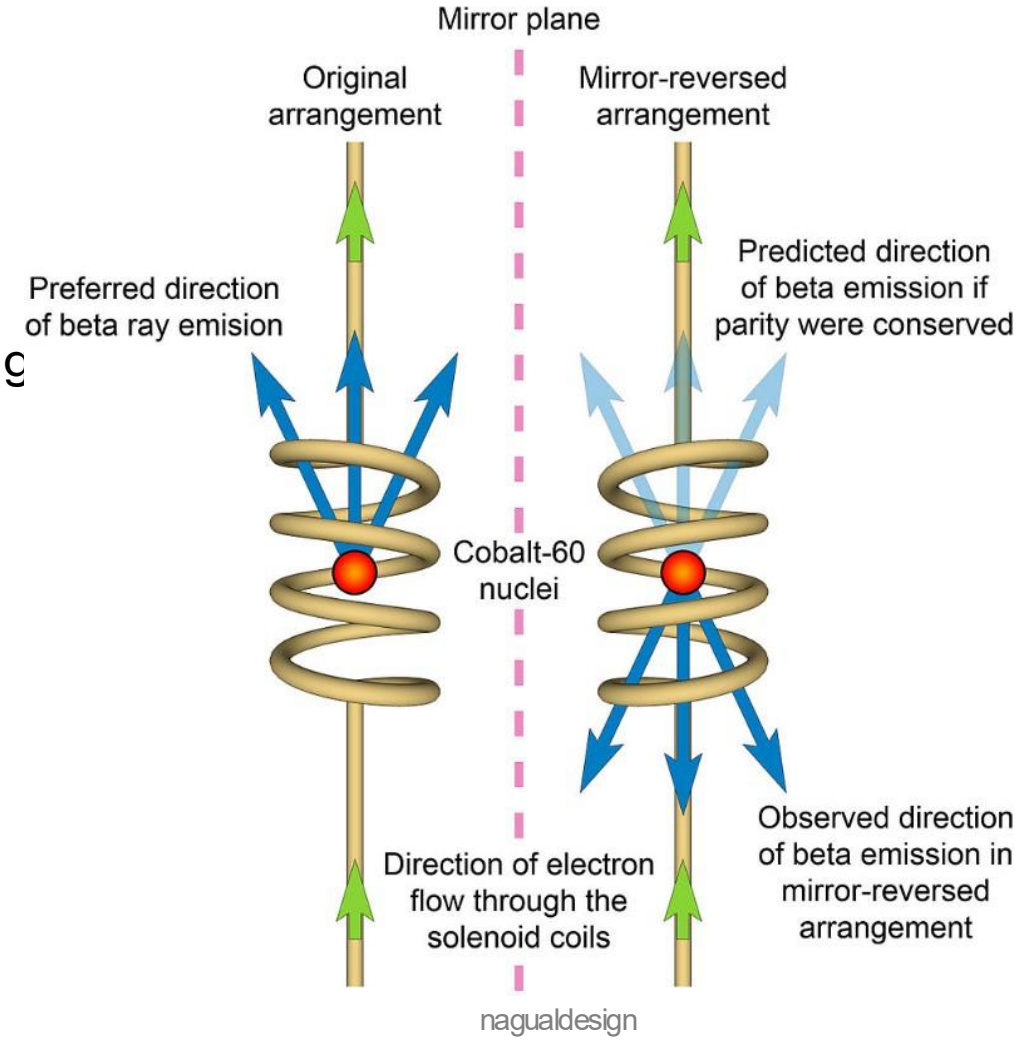
- When the decay energy is < 1.022 MeV ($2m_e c^2$) but larger than 0, the β decay of a proton-rich nucleus to its daughter must be via electron capture (EC)
- For decay energies > 1.022 MeV, $\beta^+$and EC compete.
- In EC decay, only one particle is emitted from the nucleus (the neutrino).
- Assuming a zero neutrino rest mass, the decay constant for EC can be written as

$$\lambda_{\mathrm{EC}} = \frac{\boldsymbol{g^2 \left| M_{fi} \right|^2} \mathrm{T}_{\boldsymbol{v}}^2}{\boldsymbol{2\pi^3 \hbar^7 c^3}} |\phi_{\mathrm{K}}(0)|^2$$

- The K electron wave function can be written as $\phi_{\mathrm{K}}(0) = \frac{1}{\sqrt{\pi}} \left(\frac{\mathrm{Z m_e e^2}}{4\pi\epsilon_0 \hbar^2} \right)^{3/2}$
- So, the ratio between $\beta^+$and EC goes as $\frac{\lambda_{\mathrm{EC}}}{\lambda_{\beta+}} \propto \mathrm{Z}^3$ meaning that EC is more probable for heavy nuclei

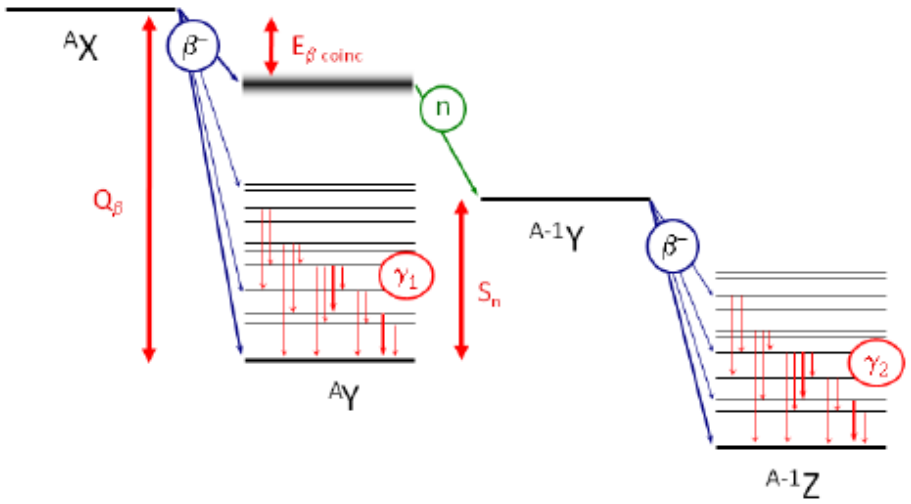
Parity Non-Conservation

- For α-decay, parity was an important consideration for selection rules
  - Transitions were only possible for which the parity change was $\Delta\pi = (-1)^{\ell}$
- For weak transitions, this is not the case
- It was demonstrated by observing the ^{60}Co β-decay angular distribution for ^{60}Co with its spin aligned along and against a magnetic field
- If parity were conserved, reflecting the spatial coordinates (by preparing ^{60}Co as spin down instead of spin up) shouldn't change the β angular distribution...but it did
- This showed weak transitions don't conserve parity
- This is the famous experiment by Madame Chien-Shiung Wu



A Few Extras

- A few topics that I did not mention are β-delayed particle emission (n, p) and statistical β-decay
- Double β decay



- HW #2 due **TODAY** at 11:59PM
- Re-submissions of HW#1 problem #7 due on Tues. Sept. 30th in class
- Office Hours on Friday at 9:30-10:30AM
  - Edwards 204
- **Next class**: Gamma Decay
- Questions?